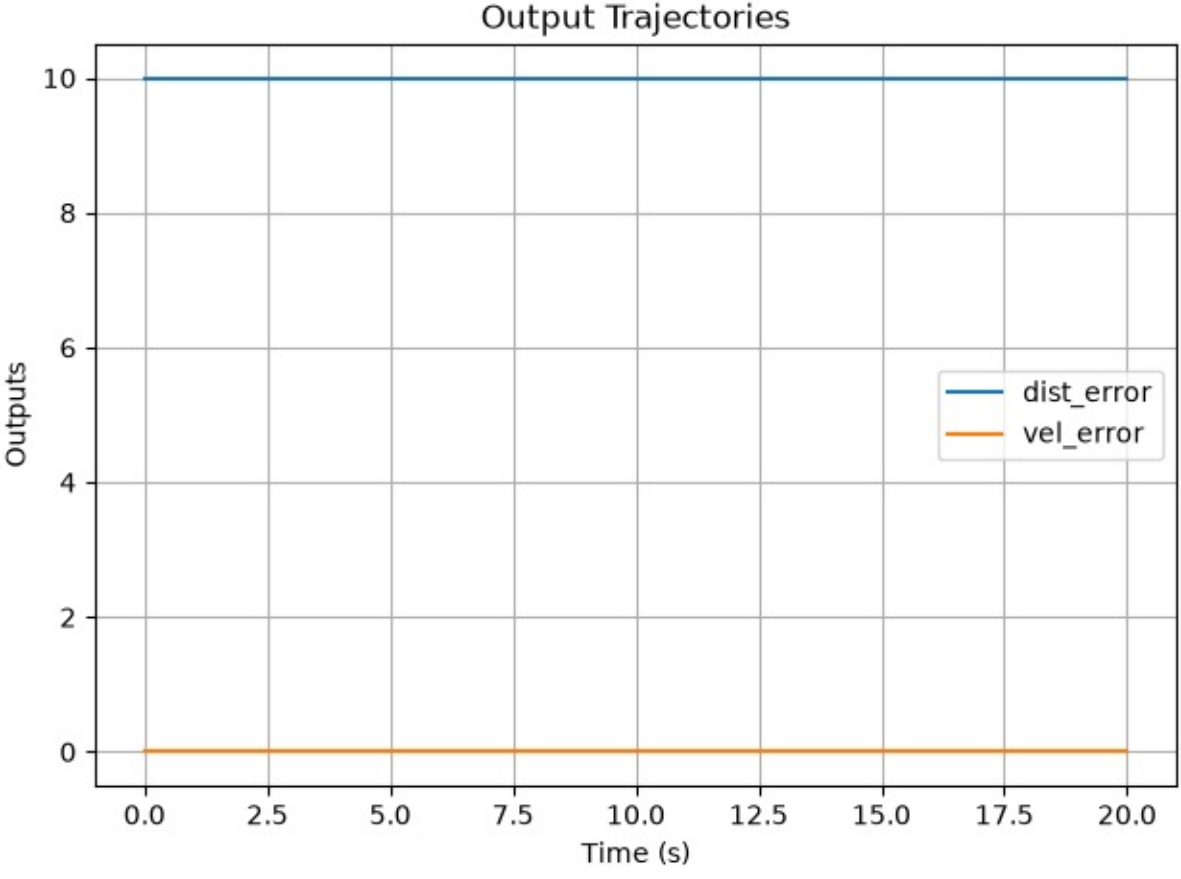
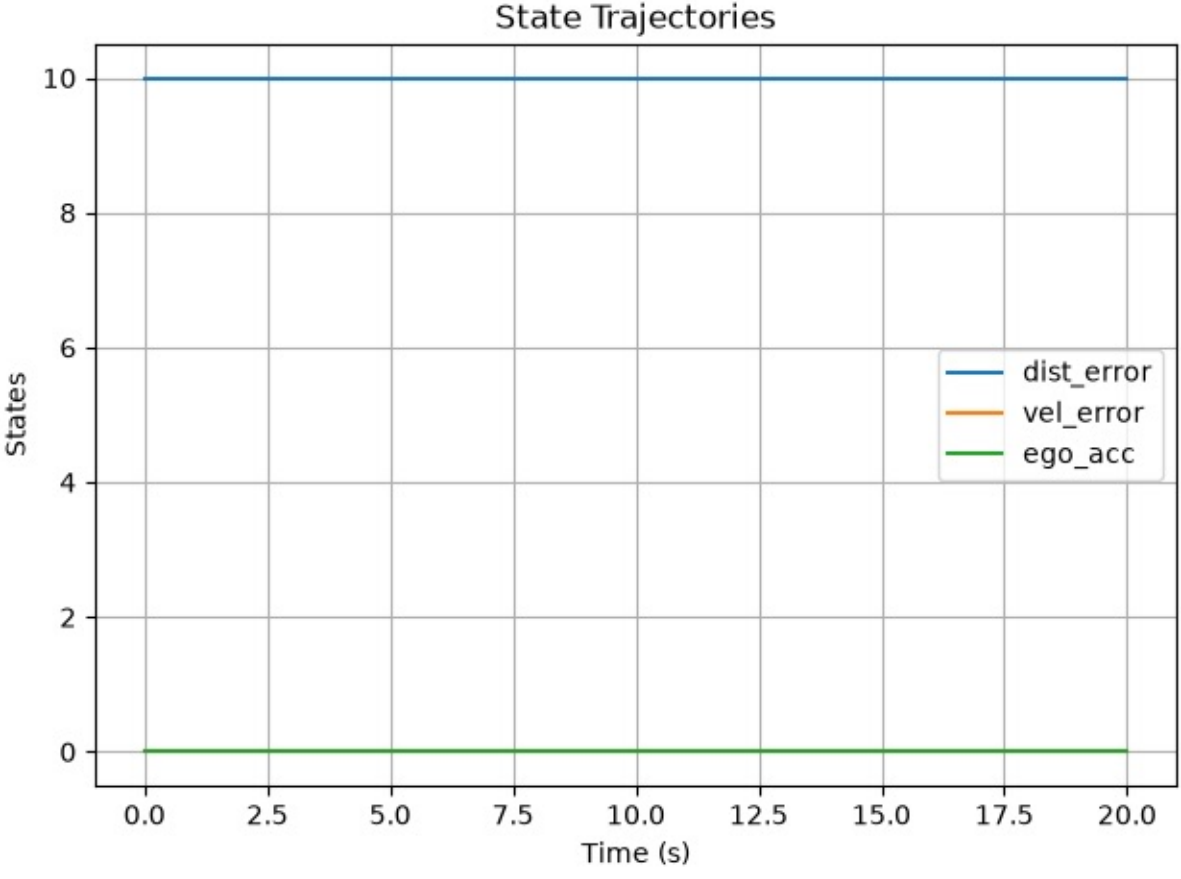
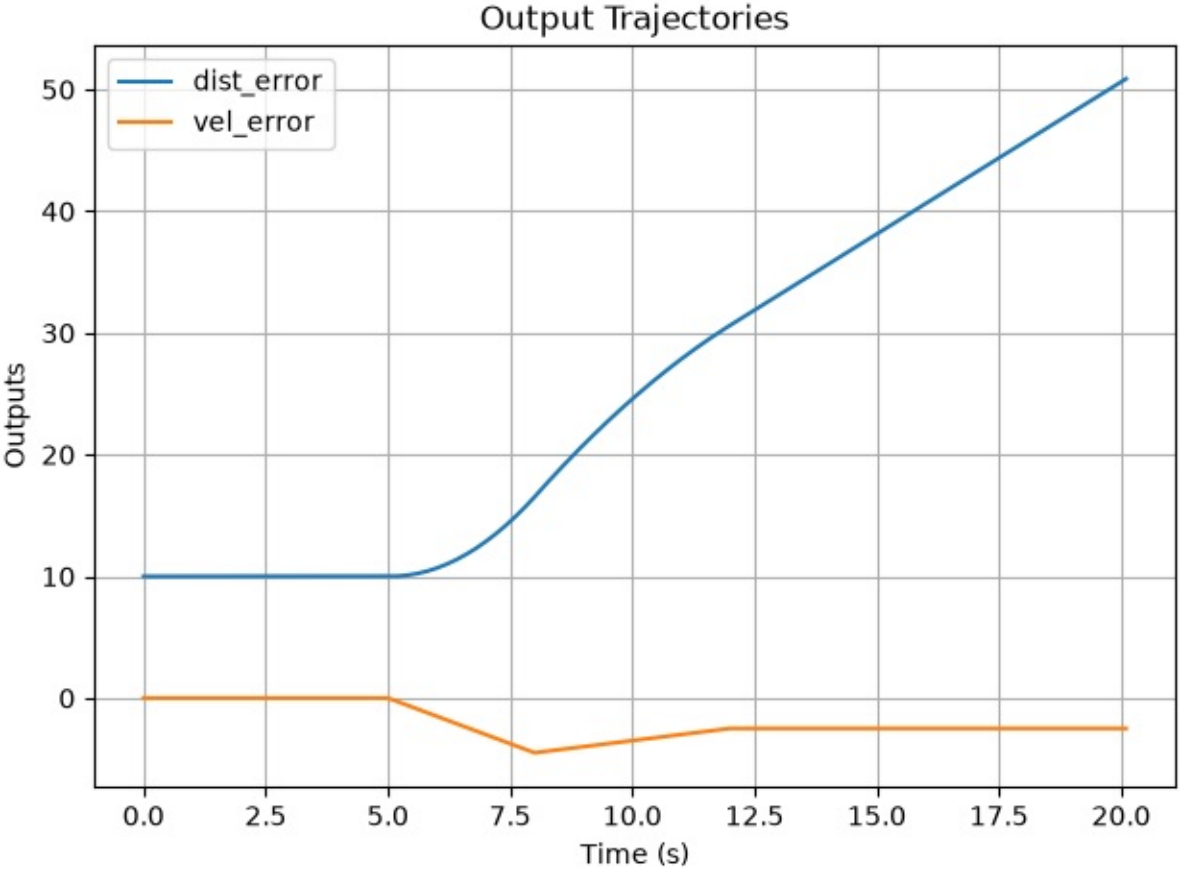
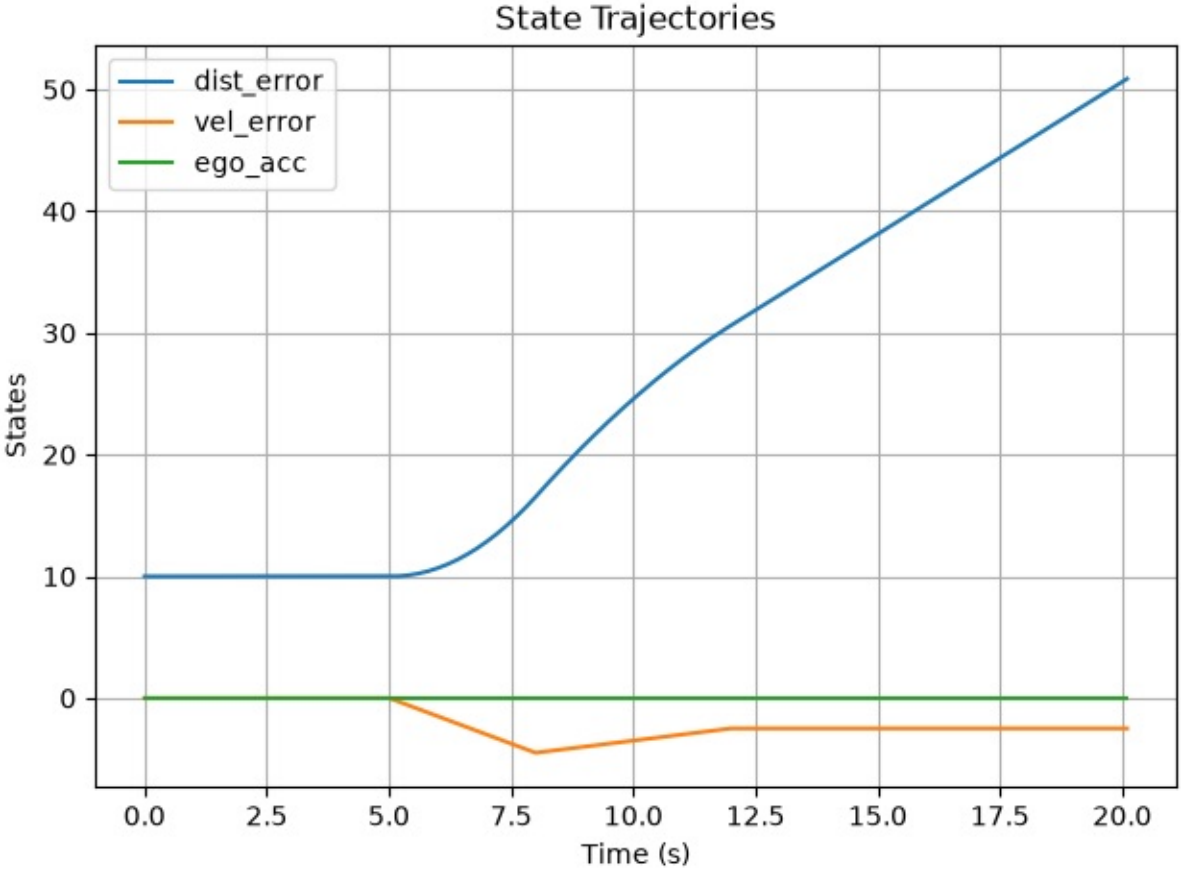

Output Trajectories
Outputs
Time (s)
dist_error
vel_error
10
8
6
4
2
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


State Trajectories
10
8
6
4
2
0
States
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)
dist_error
vel_error
ego_acc

With distance error at 10 for all time and no perturbation,
this is the sensible output of the system. Furthermore, velocity and aceleration error should be 0 as ego car is staying constant distance of 10m away from lead for all time.


Output Trajectories
dist_error
vel_error
Outputs
50
40
30
20
10
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


State Trajectories
dist_error
vel_error
ego_acc
States
50
40
30
20
10
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

Physical Interpretation:
For all time, ego car acceleration input is 0, and initial state of ego car acceleration is 0, so it makes sense that green line in 2nd plot is flat at 0.

Between 5 and 8 seconds, the lead car is decelerating at -1.5 m/s^2. Thus, the velocity of the lead car is decreasing at a constant rate. This indicates that the relative velocity from the lead car to the ego car is decreasing constantly, as the lead car velocity is decreasing but the ego car velocity stays constant. Thus the distance error increases as the distance between the lead car and ego car shrinks.

Between 8 and 12 seconds, the opposite happens. The lead car is accelerating, so the velocity difference between the lead car and ego car increases at a constant rate. Thus, ev(k) increases at a constant rate. The distance error still increases but the distance error graph changes convexity because from 5 to 8 seconds, the lead car was decelerating but from 8 to 12 seconds the lead car is accelerating.

Finally, beyond 12 seconds, the lead car is not accelerating, so the relative velocity error stays at a constant value. However, even though the lead car's velocity is now constant, the negative value of e_v(k) indicates that the lead car is slower than the ego car. Thus, the distance error is still increasing at a constant rate as the ego car "catches up" to the lead car at a constant rate.